$\sum_{n\geq 0} a_n$, $\left|\frac{a_{n+1}}{a_n}\right| \xrightarrow[n\to+\infty]{} L$: $L<1$: $\sum a_n$ CV ; $L>1$ $\sum a_n$ DV ; $L=1$ ⑤ $\longrightarrow$ $\sum 1/n : \frac{1/(n+1)}{1/n} \to 1$ pas DV ; $\sum 1/n^2 : \frac{1/(n+1)^2}{1/n^2} \to 1$ pas CV

$a_N \leq L \cdot a_{N-1} \leq L^2 a_{N-2} \leq \dots \leq L^{N-n_0} a_{n_0}$

$\sum_{n_0}^{N} a_n \leq a_{n_0} \sum_{k=n_0}^{N} L^k \xrightarrow[N\to+\infty]{} M < +\infty$.

$\Rightarrow L<1$

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a- $\sum_{n\geq 1} \frac{2n-1}{(\sqrt{2})^n}$, $\frac{(2(n+1)-1)/(\sqrt{2})^{n+1}}{(2n-1)/\sqrt{2}^n} = \frac{1}{\sqrt{2}} \cdot \frac{2n+1}{2n-1} \xrightarrow[n\to+\infty]{} \frac{1}{\sqrt{2}} < 1$.

$\left(\frac{n+1}{n}\right)^{-n} = \left(1+1/n\right)^{-n} = \left(\left(1+1/n\right)^{n}\right)^{-1} = \frac{1}{(1+1/n)^n}$

entonces : $\sum \frac{2n-1}{(\sqrt{2})^n}$ CV.

b- $\sum_{n\geq 1} \frac{n!}{n^n}$, $\frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} = \underbrace{(n+1)}_{} \times \frac{1}{(n+1)} \times \left(\frac{n}{n+1}\right)^n = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n} = 1/e = e^{-1}$

$(n+1)! = (n+1)n!$

$(n+1)^{n+1} = (n+1)^n \times (n+1)$

$\Rightarrow \sum_{n\geq 1} n!/n^n$ CV !

$\exp\left(\log\left(\frac{n}{n+1}\right) \times n\right) = \exp\left(\underbrace{\log(1+1/n)}_{1/n + o(1/n)} \times n\right) \to \exp(1)$

$n \times o(1/n) = o(1)$

$\frac{a_n}{\frac{1}{n}} \to 0$

$e \simeq 2{,}78$ $1/e \leq 1$

5- C : Cauchy : $\sqrt[n]{a_n} \xrightarrow[n\to+\infty]{} L$: <1 CV ; >1 DV ; $=1$?

$\sum_{n\geq 1} \frac{2n-1}{(\sqrt{2})^n}$: $\sqrt[n]{\frac{2n-1}{\sqrt{2}^n}} = \frac{\sqrt[n]{2n-1}}{\sqrt{2}}$, $\sqrt[n]{n} = n^{(1/n)} = \exp\left(\ln(n) \times \frac{1}{n}\right)$ log. $\frac{\ln(n)}{n} \to 0$

$\exp(0) = 1$.

$\leq \frac{\sqrt[n]{2n}}{\sqrt{2}}$

$\to 1/\sqrt{2} < 1$

$2n < 2n-2 = 2(n-1)$

$\sqrt[n]{\lambda n} = \sqrt[n]{\lambda} \cdot \sqrt[n]{n} \simeq \sqrt[n]{n}$

$\Rightarrow \sqrt[n]{a}$

$a_n = o(b_n)^{1/n^{1/2}}$, $:= \frac{a_n}{b_n} \to 0$

$\frac{\ln(n+1)}{\ln(n)} \cdot \frac{n}{n+1}$

$\ln(1+1/n) \to 1$ ($\to h$, $\ln(1)=0$)

$\Rightarrow \sum_{n\geq 1} \frac{2n-1}{(\sqrt{2})^n}$ CV.

$\sqrt[n]{\lambda} = \exp(\lambda \cdot 1/n) \to \exp(0) = 1$.